

GHOST

OF TSUSHIMA

Procedural Grass



Concept art



Concept art



Concept art

Dense foliage

Many types of plants

Grass card method



+ Texture

Cheap

Overdraw

Low density feeling

Meh graphics



1M grass blades

83k rendered

2.5 ms (PS4)







Render Tile = 512 x 512

Height map

Ground Texture

Grass type map

Grass Height

...

Render Tile = 512 x 512

Height map

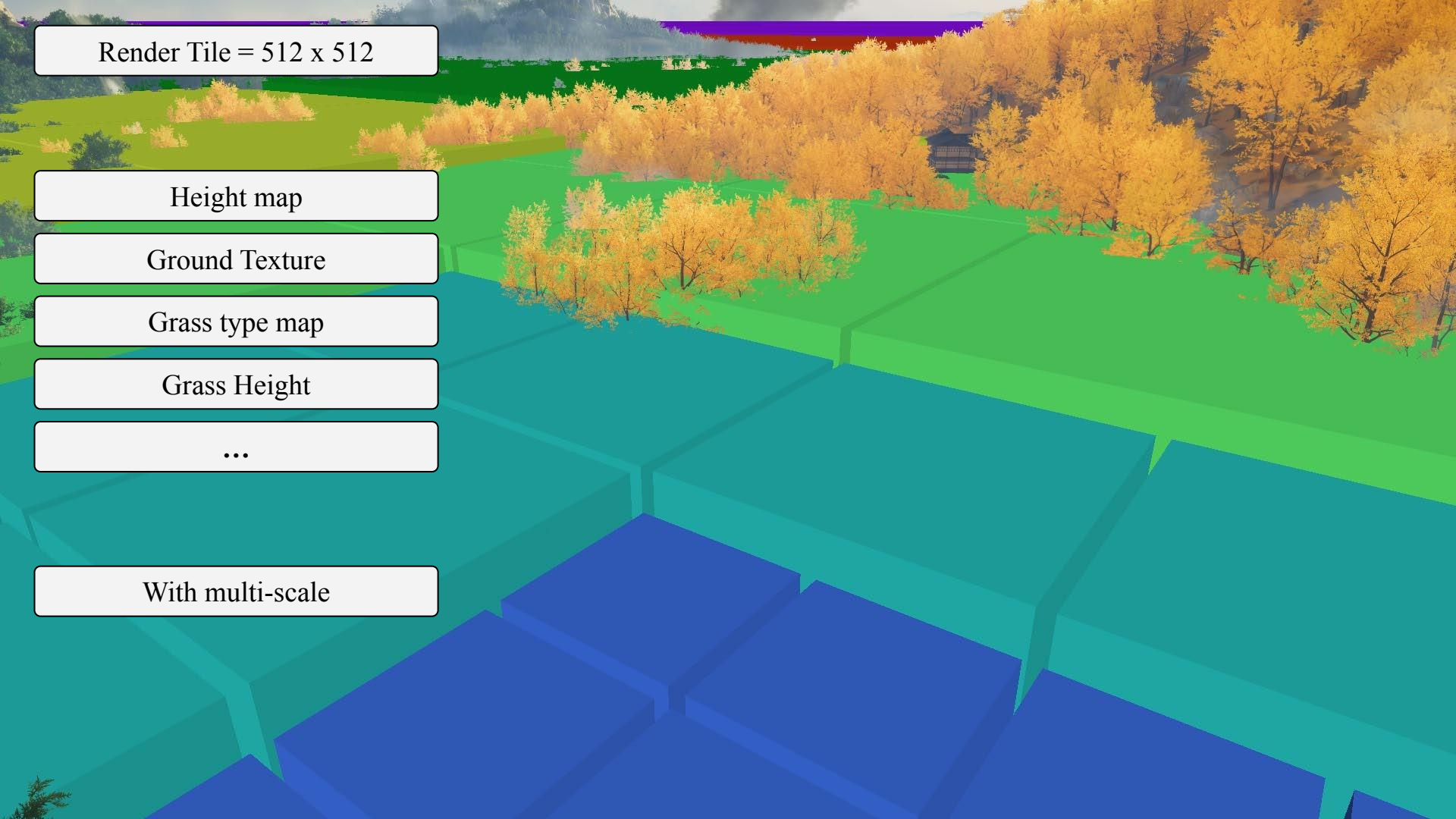
Ground Texture

Grass type map

Grass Height

...

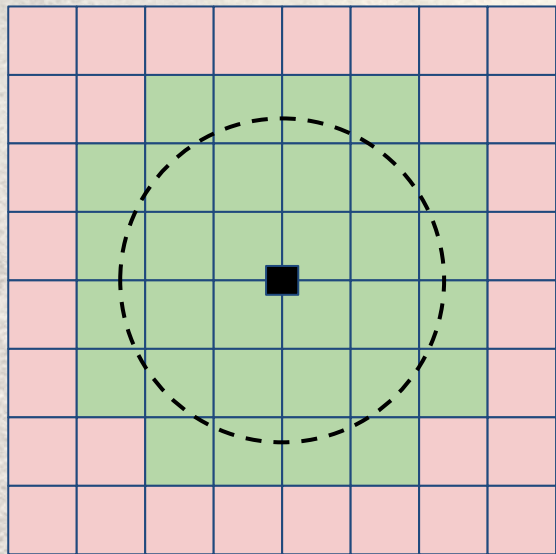
With multi-scale



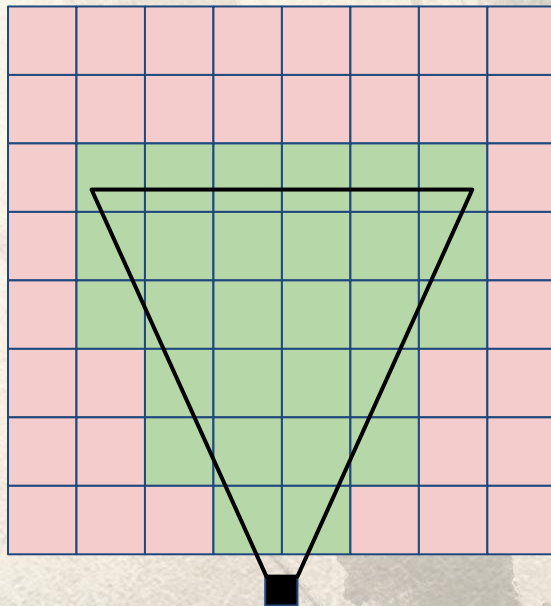
Placing a Grass Blade

- 1 - Turn lane ID into position on tile grid, jitter (rng offset)
- 2 - Distance culling and frustum culling
- 3 - Determine grass type and height from texture at position
- 4 - Drop lanes that don't have grass
- 5 - Occlusion culling

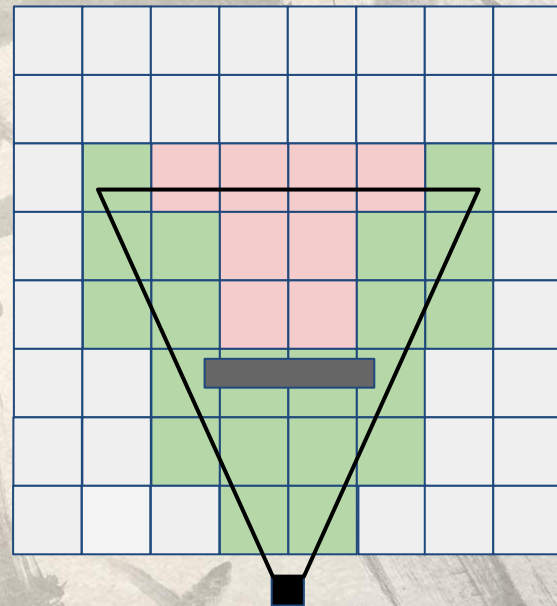
Distance Culling

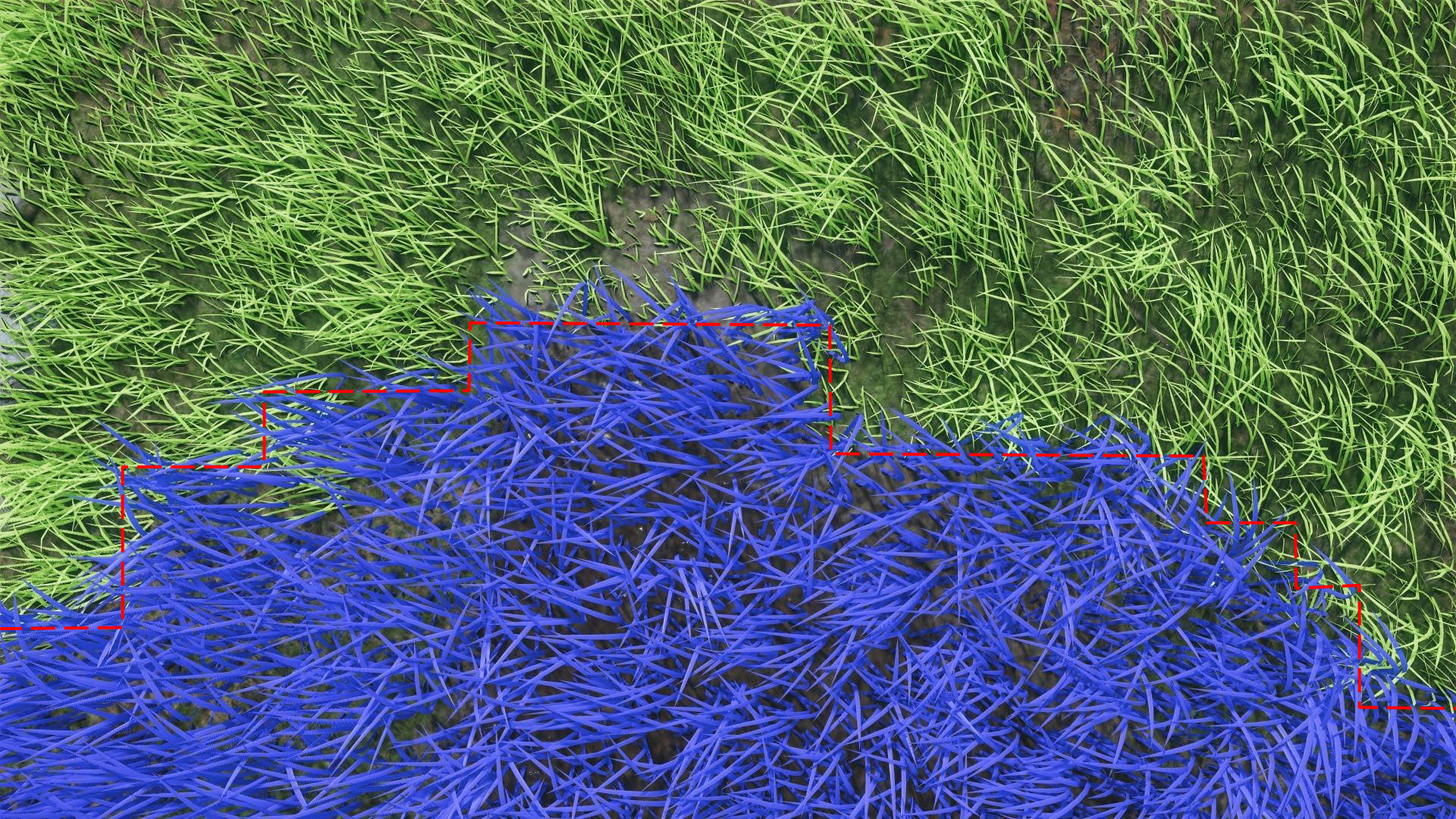


Frustum Culling



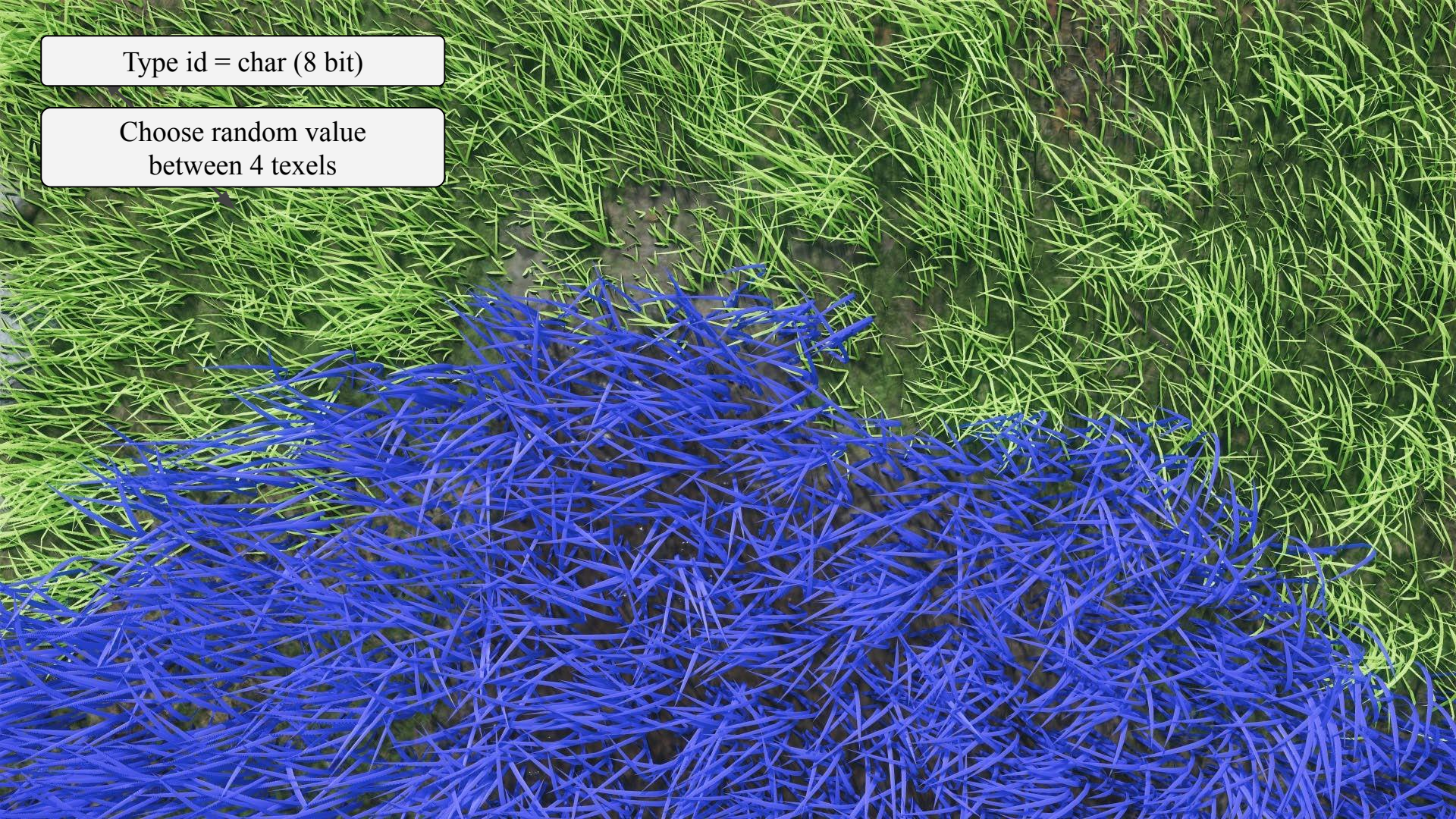
Occlusion Culling





Type id = char (8 bit)

Choose random value
between 4 texels



- Position (vec3)
- Facing (vec2)
- Per-blade Hash
- (Wind strength at position)

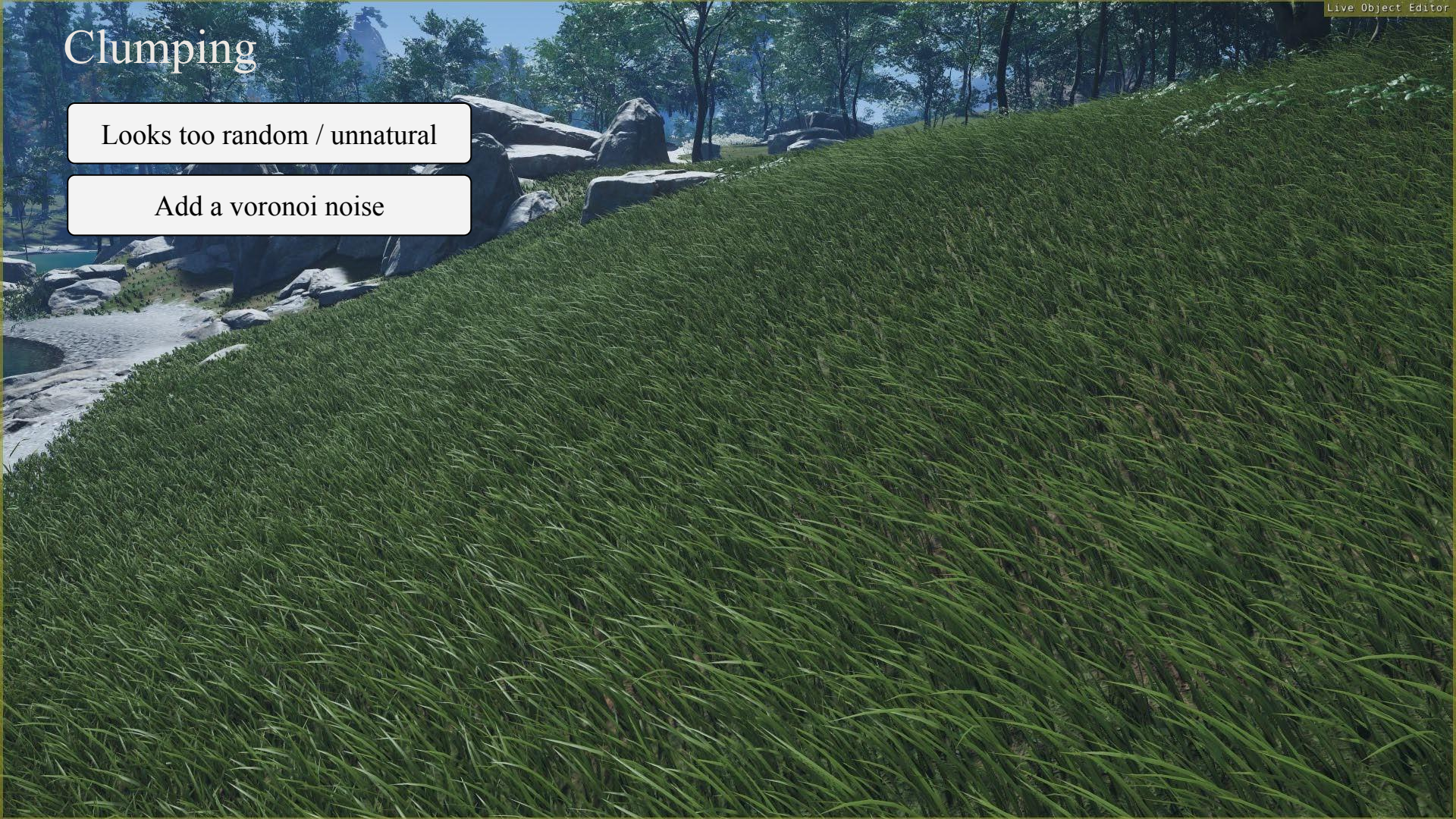
Artist control

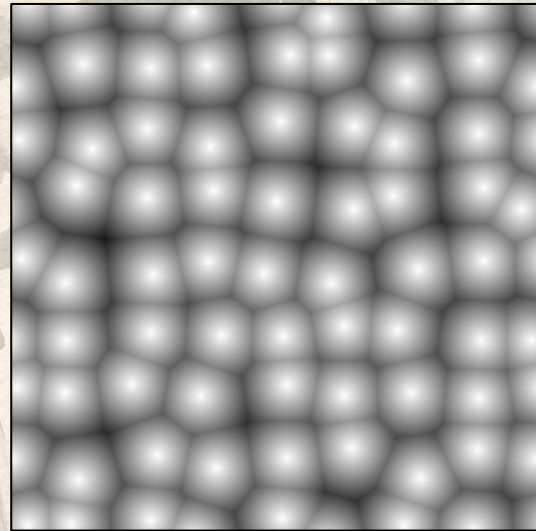
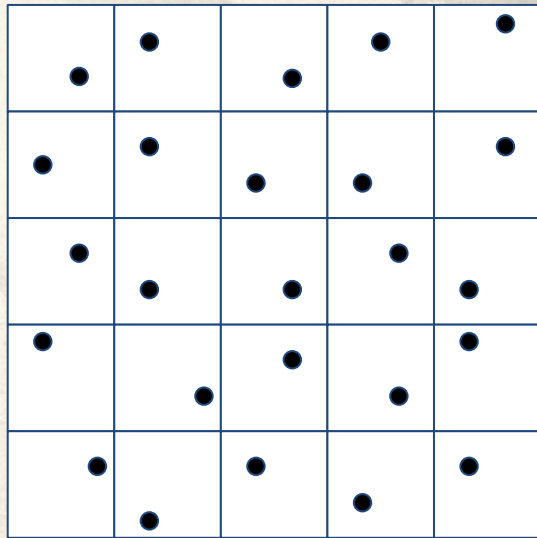
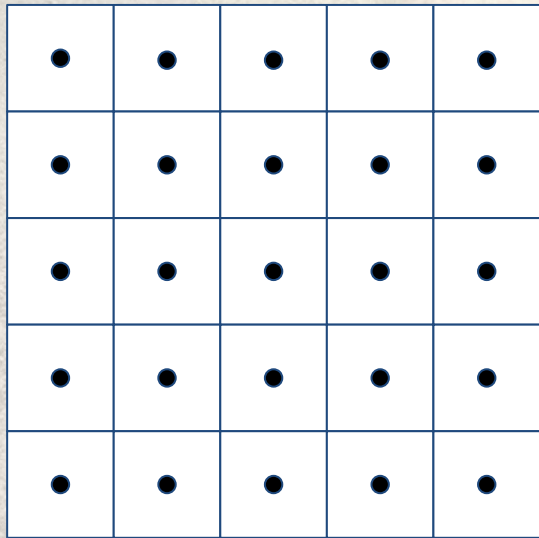
- Clump facing (2 floats)
- Clump color
- Grass Type
- Height
- Width
- Tilt
- Bend
- Side Curve

Clumping

Looks too random / unnatural

Add a voronoi noise





Clumping

Change height



Clumping

Direction per clump



Clumping

add offset



Clumping

direction out of clump



Clumping

all together

New

Old



High LOD
15 vertices



Low LOD
7 vertices

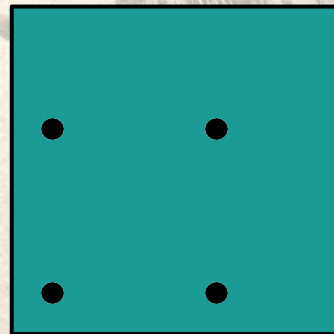
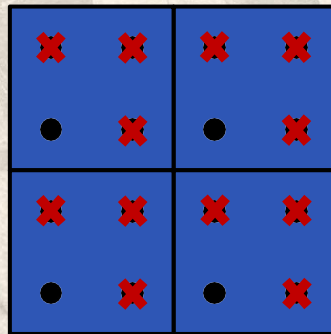


can choose the distribution of vertices

Transition Low $\Rightarrow$ high

Linear interpolation of geometry

remove blade before transition



High LOD
15 vertices



Low LOD
7 vertices



When grass is short

Much denser

Low LOD is weird ... that's all

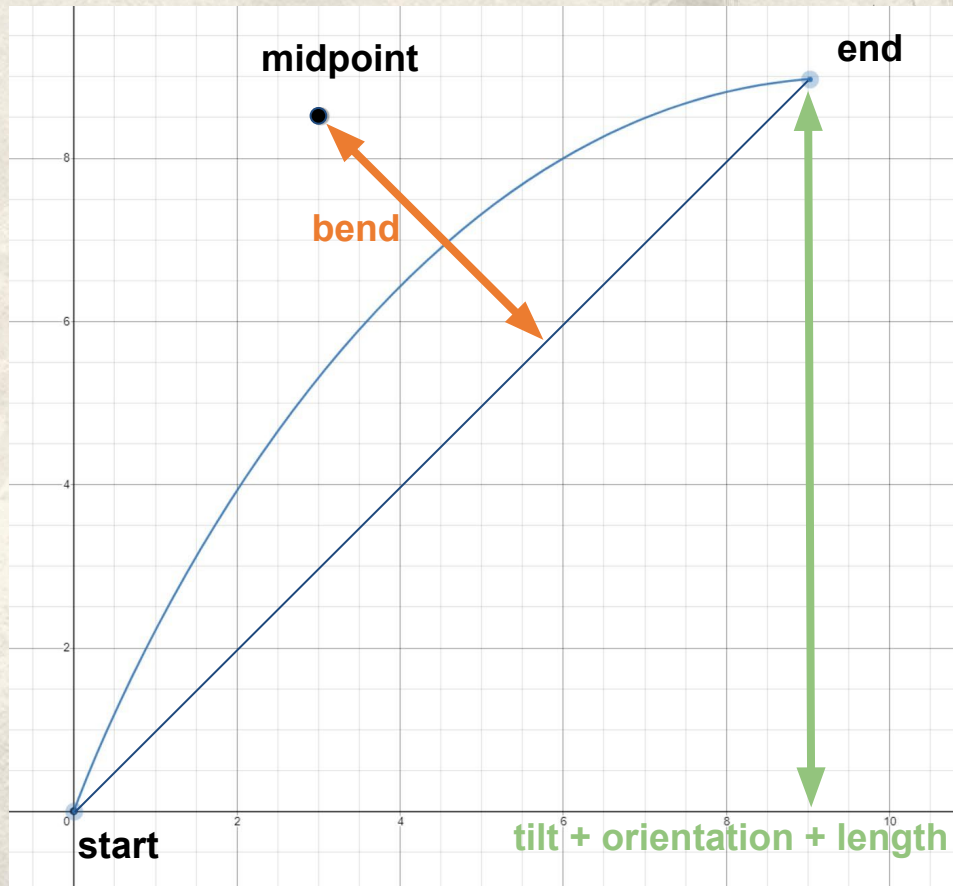
Cubic Bezier curve

- Easy position + normal evaluation
- Control point changes blade shape

Blade Generation

- Evaluate Bezier curve
- Get normal orthogonal to facing
- Step vertex in width direction
- Evaluate Bezier derivative curve



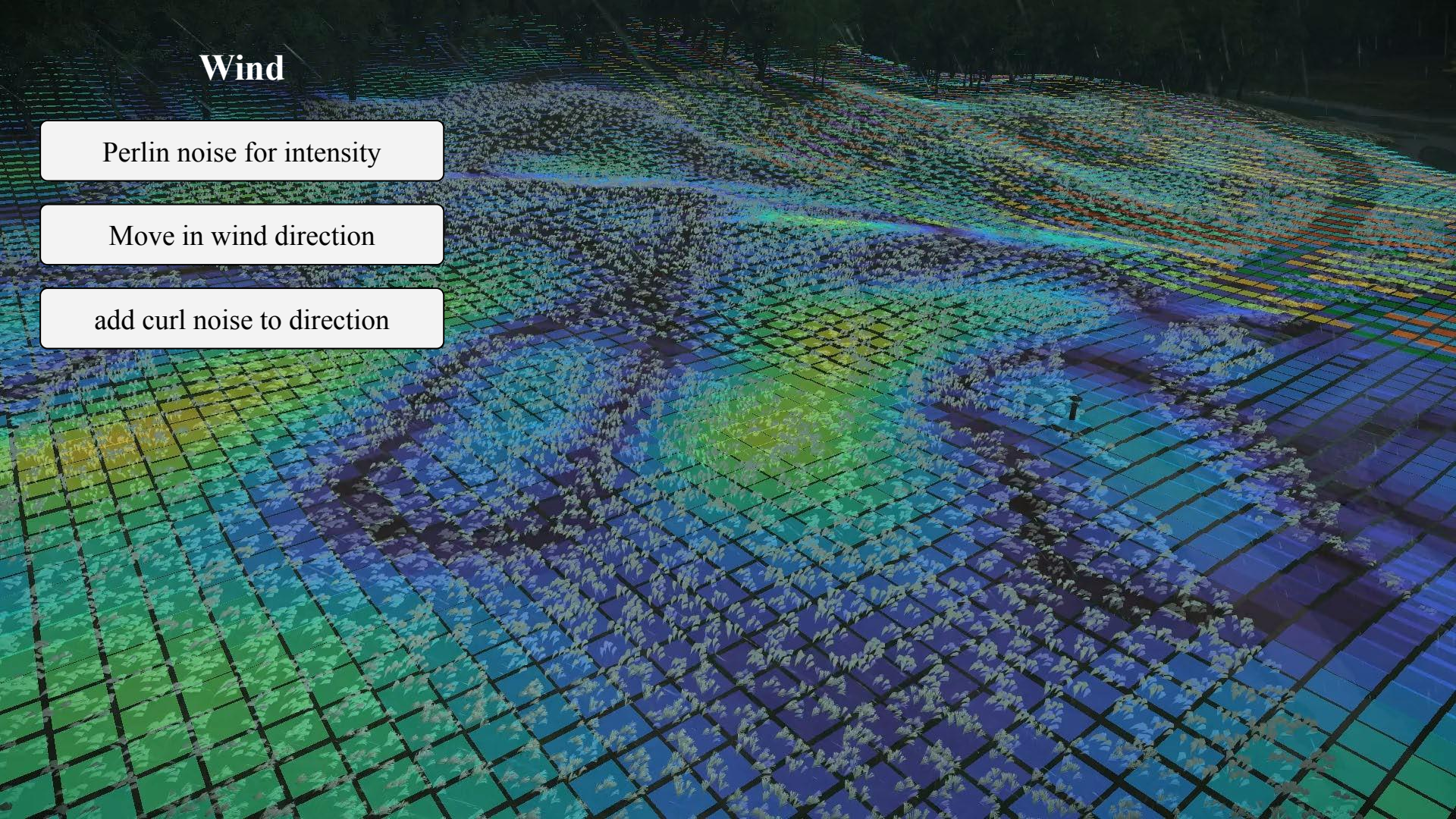


Wind

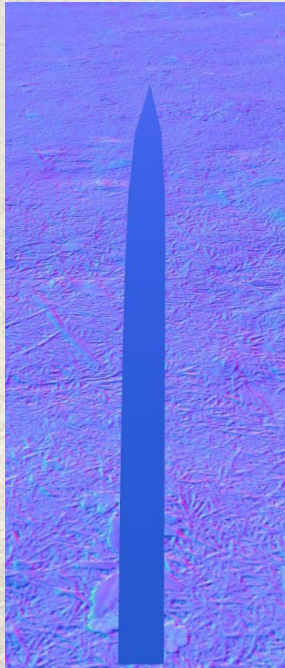
Perlin noise for intensity

Move in wind direction

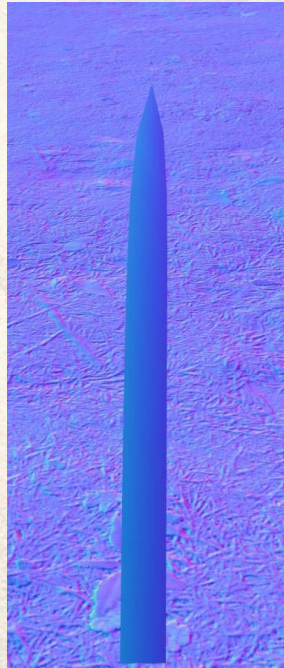
add curl noise to direction



Flat normals



Rounded normals



Can deduce rounded normal

Controlled by “Side Curve” ?

Facing bias

seems empty

orthogonal blades are thin



Facing bias

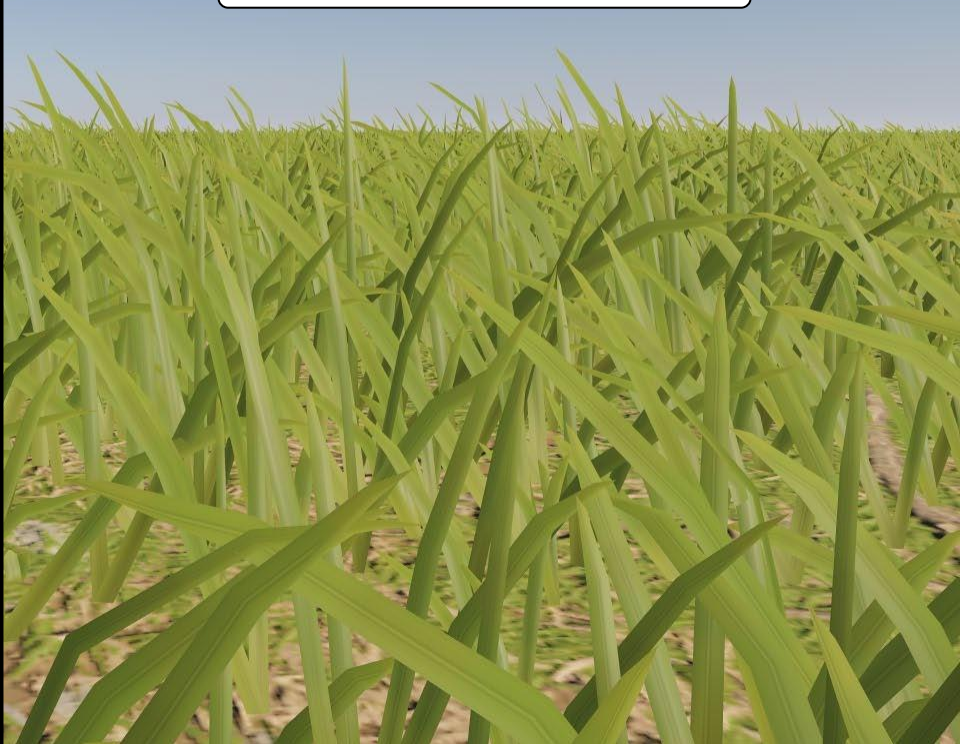
seems empty

orthogonal blades are thin



Add shift in view space

like billboard ?



Noise Problem

glossy / specular material

normal variation

lot of aliasing when far



Noise Problem

glossy / specular material

normal variation

lot of aliasing when far



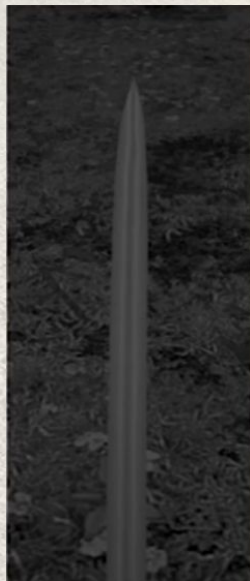
Update with distance

lerp normal to global normal

reduce gloss



gloss



1D texture

vein



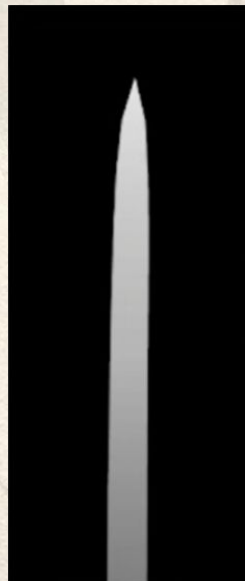
1D texture

diffuse



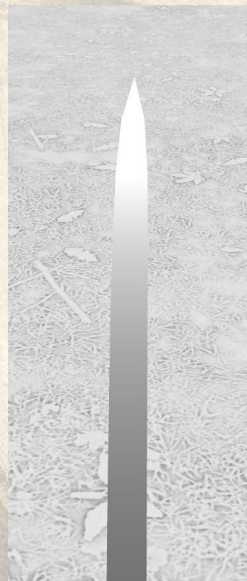
2D texture

translucency



constant

**ambient
occlusion**



constant



final





Artists can place assets to add diversity



Many applications



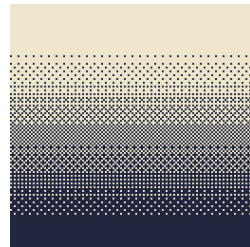
Far => replaced by texture

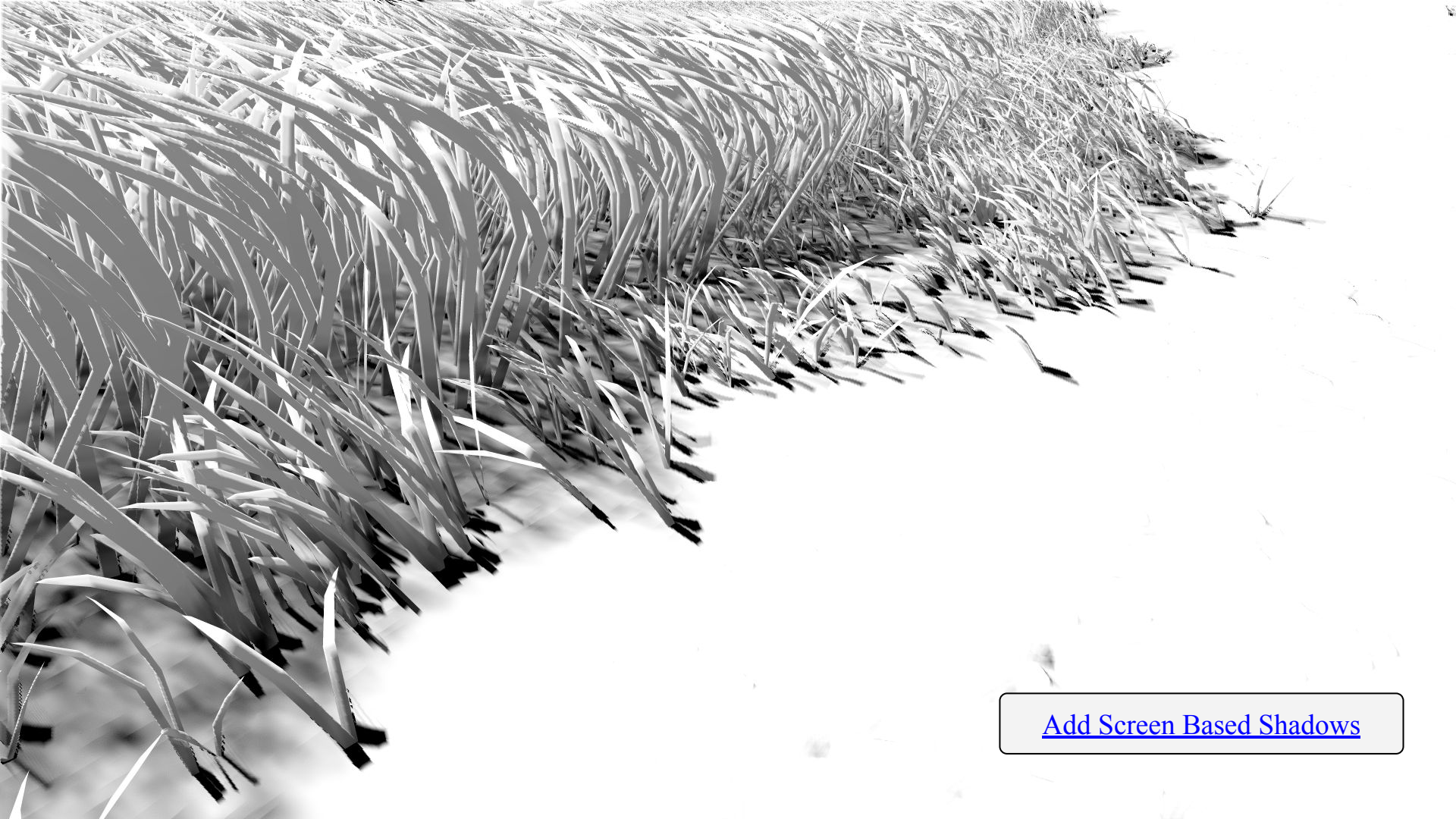




Deform terrain to match grass height

add dithered pattern on shadow map

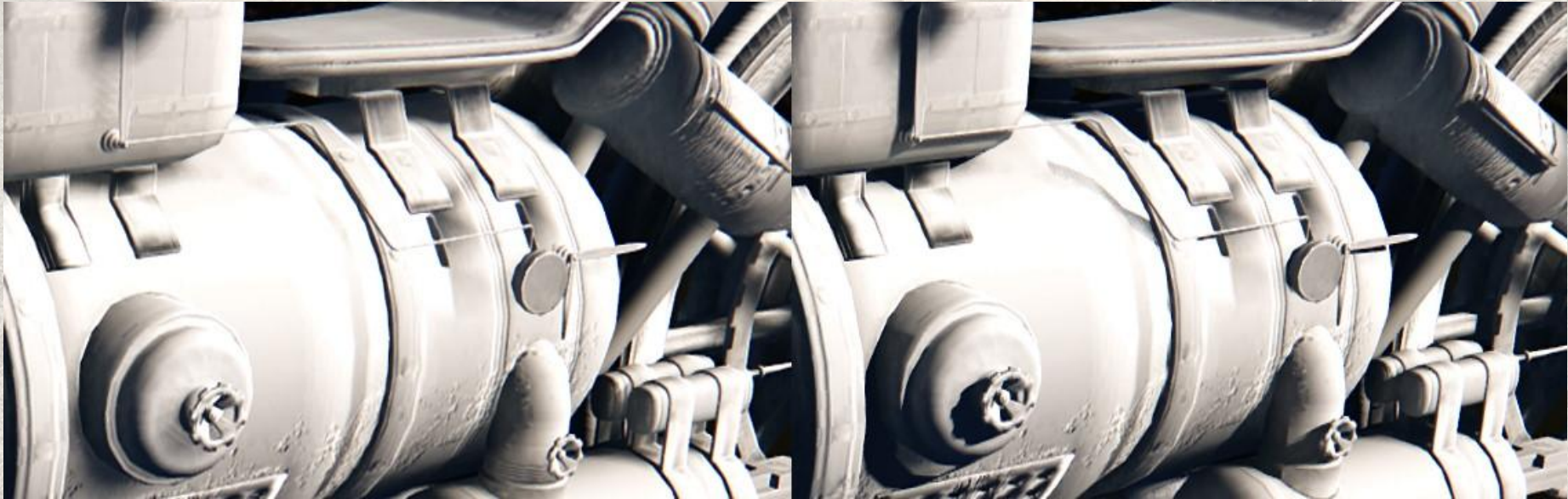




[Add Screen Based Shadows](#)

Shadow Map

+ Screen Space Shadow



Add fine details to shadow + Cheap

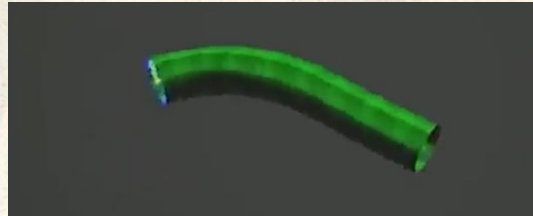
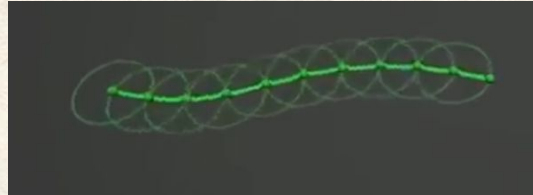




Conflict between terrain and assets



Limited with foliage type



Flexible particle system

Grass, Tentacles, Destructible Terrain, Fog, etc.